

Java Foundations: Language Skills for System Design & Production Engineering

1. Data Structures & Algorithms

- 1 Arrays, Linked Lists, Stacks, Queues
- 2 Hashing and HashMaps (collision handling, resizing)
- 3 Trees (BST, AVL), Graphs (BFS, DFS)
- 4 Sorting & Searching algorithms
- 5 Time & Space complexity analysis
- 6 Problem-solving patterns (sliding window, recursion, dynamic programming)

2. Concurrency & Multithreading

- 1 Thread lifecycle and thread pools
- 2 Synchronization (synchronized, locks, volatile)
- 3 ExecutorService and CompletableFuture
- 4 Deadlocks, race conditions, and thread safety
- 5 Concurrent collections (ConcurrentHashMap)
- 6 Parallel processing and ForkJoin framework

3. Networking & I/O (Blocking & Non-blocking)

- 1 Java I/O streams (InputStream, OutputStream)
- 2 File handling and serialization
- 3 Sockets and basic networking
- 4 NIO (Channels, Buffers, Selectors)
- 5 Non-blocking and reactive programming concepts
- 6 Backpressure and stream processing basics

4. Object-Oriented Programming & Design Principles

- 1 Encapsulation, Inheritance, Polymorphism, Abstraction
- 2 SOLID principles
- 3 Common design patterns (Factory, Singleton, Strategy, Observer)
- 4 Clean code practices and modular design

- 5 Immutability and object lifecycle management
- 6 Code reusability and maintainability